

Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Evidence from the Universe of Chinese Customs Transactions, 2000–2015*

Edoardo Vitella

University of Trento & Free University of Bozen

July 2026 — **preliminary draft**

Abstract

Environmental provisions (EPs) are now ubiquitous in preferential trade agreements (PTAs), yet evidence on their trade effects remains scarce and is based on aggregate data. I study whether the EPs contained in the PTAs signed by China between 2000 and 2015 changed the *composition* of Chinese exports — shifting firms’ export baskets toward environmental (green) goods and away from pollution-intensive (dirty) goods — using transaction-level customs data covering 45.8 million firm–product–destination–year observations. Identification exploits a triple-difference design: within a given firm–destination–year, I compare green and dirty products to neutral ones as EP depth varies across destinations and over time, absorbing the confounding effect of the agreement itself through firm–destination–year fixed effects. I find a null on the green margin, stable across six estimation designs (full panel, collapsed panel, and four control-group subsamples), surviving wild cluster bootstrap and permutation inference. Bootstrap bounds rule out effects larger than roughly one quarter of the benchmark estimate in the aggregate literature, while leaving smaller effects undetermined. The dirty margin is a textbook false positive: a negative effect on dirty exports that is overwhelming asymptotically ($p < 0.001$) survives neither the wild cluster bootstrap ($p = 0.07$) nor permutation inference ($p = 0.23$), a coarser aggregation of the same test reverses its sign, and a leave-one-out exercise shows it is carried by a single destination (Australia); it is best read as a descriptive pattern, not an identified effect. The central reading is not that environmental provisions are inert, but that Chinese agreements bundle provisions

*First draft. Comments welcome. All remaining errors are my own.

so tightly that the two World Bank sub-indices with a direct trade mechanism (green-goods liberalization, binding standards) are perfectly collinear in-sample and non-zero in only three treated country-years: mechanism-bearing content is essentially absent from Chinese PTAs of this period, and the data cannot separately identify what it would do. This is consistent with, not a contrast to, Brandi et al. (2020): their effects are driven by explicitly trade-restrictive or trade-liberalizing clauses, and China’s 2000–2015 agreements — dominated by cooperation language — contain almost none of that content. The result is compatible with Abman et al. (2024) on the same logic: specific, enforceable provisions can move outcomes; generic ones do not.

Keywords: trade agreements, environmental provisions, export composition, China.

JEL: F13, F14, F18, Q56.

1 Introduction

Nearly every trade agreement signed in the last two decades contains environmental language. The TREND database (Morin et al., 2018) codes roughly 300 distinct types of environmental provisions (EPs) across more than 700 preferential trade agreements (PTAs), and the depth of this content has grown steadily. Whether these provisions have real effects — on trade flows or on environmental outcomes — is a question that the authoritative survey of the trade–environment literature flags as open (Copeland et al., 2022). Two recent papers provide the sharpest evidence to date. Brandi et al. (2020) combine TREND with aggregate bilateral trade flows and find that trade-restrictive EPs reduce the dirty share, and liberal EPs raise the green share, of developing-country exports. Abman et al. (2024) show that forest- and biodiversity-specific provisions fully offset the deforestation that otherwise follows entry into force of a regional trade agreement.

This paper brings the question to the world’s largest exporter with the most granular data available: the universe of Chinese customs transactions, observed at the firm–product(HS6)–destination–year level (49.2 million observations, roughly 462,000 exporting firms). Between 2000 and 2015 China signed around 14 PTAs — covering 25 destination economies — whose environmental content ranges from essentially none (the Bangkok Agreement) to substantial chapters (New Zealand 2008, Switzerland 2014, Korea 2015). I ask whether deeper environmental content shifted the *composition* of Chinese exports toward green goods and away from dirty goods, relative to neutral products, within the same firm, destination, and year.

The focus on composition, rather than on export volumes, is dictated by identification. EP depth in Chinese agreements varies at the destination–year level, enters in force together with the agreement itself, and afterwards almost never changes (only three destinations exhibit within-country variation). Any level effect of EP depth on trade is therefore observationally equivalent to the effect of the agreement, and of agreement depth in general — a collinearity that no fixed-effects structure can undo. I document this formally with a saturation ladder: coefficients on EP depth shrink monotonically to a precise zero as fixed effects absorb the level of the trade relationship, the signature of selection into agreements rather than of a causal effect (Bertrand et al., 2004; Abadie et al., 2023). The composition question, by contrast, is identified: within a firm–destination–year cell, the agreement affects green, dirty, and neutral products symmetrically, so their *differential* response to EP depth isolates the environmental content. The design follows the logic of the classic industry- $\times$ -country interaction of Rajan & Zingales (1998), and mirrors the contrast that Abman et al. (2024) exploit between agreements that do and do not contain specific environmental content, holding the existence of an agreement fixed. The analogy is structural, not literal: Rajan & Zingales’s industries

are sorted by a technological trait (external finance dependence) that is fixed across countries, whereas green and dirty status here is a *policy*-defined product classification (OECD and Mani–Wheeler lists), a distinction worth flagging even though it does not vary with the agreement and so cannot itself confound identification.

The answer is a null on both margins, bounded but not razor-sharp. Six independent estimation designs — the full firm-level panel, a product–destination–year collapsed panel, and four control-group subsamples that probe the stability of the coefficient across comparison sets — return an $EP \times \text{green}$ coefficient between -0.0009 and -0.0046 (in log points per provision), never remotely significant, with standard errors around 0.003 – 0.007 . The dynamic version shows flat differential pre-trends and no break at entry into force. Inference is conducted at three levels — destination-clustered asymptotics, wild cluster bootstrap (Cameron et al., 2008), and permutation tests that reshuffle entire EP profiles across treated destinations (Fisher, 1935; Bertrand et al., 2004) — because only 23 destinations (and effectively 14 agreements) are treated. A negative coefficient on $EP \times \text{dirty}$ (asymptotic $p < 0.001$ in the collapsed panel) illustrates exactly why this discipline matters: its wild-bootstrap p-value is 0.07 , permutation inference on the exact estimating equation gives $p = 0.23$, a coarser aggregate version of the same permutation flips its sign ($p = 0.49$), the TREND index never confirms it, and a leave-one-out exercise shows it is carried by a single destination (Australia, $p = 0.24$ once excluded); South Korea, the destination whose removal previously killed the result, now leaves it borderline ($p = 0.09$). No form of robust inference sustains it. The joint F-test on all four composition interactions in the full firm-level panel is $p = 0.31$ (WB index) and $p = 0.71$ (TREND index).

The null is informative rather than disappointing, and its central reading is about *content*, not about environmental provisions in general. Decomposing the indices reveals that Chinese EPs of the period bundle provisions so tightly that the two World Bank sub-indices with a direct trade mechanism — green-goods liberalization and environmental standards — are *perfectly* collinear in-sample ($\rho = 1.00$) and non-zero in only three treated country-years; most environmental content is soft cooperation language with no trade mechanism at all. Where Brandi et al. (2020) find effects, they are driven by explicitly trade-restrictive or trade-liberalizing clauses; the Chinese agreements of 2000–2015 contained almost none of either. Read this way, the null is not a contrast with Brandi et al. (2020) but what their own heterogeneity predicts for agreements of this composition — and, on the same logic, compatible with Abman et al. (2024)’s finding that specific, enforceable provisions do move outcomes: China’s chapters were neither. Two further facts reinforce this reading. First, the estimate is precise: obtained on the near-universe of transactions of the world’s dominant exporter, it firmly rejects economically meaningful composition effects (say, half the magni-

tudes in Brandi et al., 2020), so the null is not merely a failure to reject. Second, it is stable across control groups — an artifact of the comparison set would move as the comparison set moves, and it does not.

This paper contributes to three literatures: the trade effects of PTA content (Hofmann et al., 2017; Dür et al., 2014; Neri-Lainé et al., 2023; Baccini et al., 2017), the trade–environment nexus (Cherniwchan, 2017; Shapiro, 2021; Dechezleprêtre & Sato, 2017), and the effectiveness of environmental clauses in trade agreements (Brandi et al., 2020; Abman et al., 2024; Morin et al., 2018; Baghdadi et al., 2013). Methodologically, it offers a template for honest inference when treatment varies across few clusters (Cameron et al., 2008; Abadie et al., 2023) and for control-group stability analysis in triple-difference designs.

The remainder of the paper proceeds as follows. Section 2 describes the agreements and the data. Section 3 presents the empirical strategy. Section 4 reports the main results, Section 5 the robustness battery, and Section 6 concludes.

2 Background and data

2.1 Chinese PTAs and their environmental content

Between 2000 and 2015, PTAs signed by China and entering into force cover 25 destination economies. Table 1 lists them. Three features of this treatment landscape discipline everything that follows. First, the ASEAN agreement (2005) covers eleven destinations with identical values of every EP index: the effective number of agreements is roughly 14, not 25. Second, EP content is fixed at signature: only three destinations (notably South Korea, which moved from the shallow Bangkok Agreement to the deep 2015 FTA) display within-country variation over time, and in no case does depth ever fall — there is no treatment reversal in the sample, so estimators built for treatment switching on and off are not required here. Third, Hong Kong and Macao (CEPA, 2003) are entrepôt economies whose recorded flows are dominated by re-exports; they account for half of the export value to treated destinations and are excluded from the main specification (robustness with inclusion reported below). Following Feenstra & Hanson (2004), a large share of “Chinese exports” to Hong Kong are in practice intermediated and re-exported to third countries at a substantial markup, so treating Hong Kong–Macao as a genuine final destination would misattribute third-country demand shocks to the bilateral relationship the fixed effects are meant to absorb.

Table 2 summarizes the timing structure directly: entry is concentrated in two large cohorts (2002 and the 2005 ASEAN accession) and then trickles in one destination at a time in most later years, so the smallest cohorts contain a single destination (Chile 2006,

Table 1: Treatment map: Chinese PTAs in force, 2000–2015

Agreement (partners)	In force	Notes
Bangkok Agr./APTA (BGD, IND, KOR, LAO, LKA)	2002/2005	minimal EP content
ASEAN–China (10 members + Timor-Leste, see note)	2005	one agreement, 11 destinations
Chile	2006	
Pakistan	2007	
New Zealand	2008	first substantial env. chapter
Singapore	2009	also ASEAN member
Peru	2010	
Costa Rica	2011	
Iceland; Switzerland	2014	richer environmental content
Australia; South Korea	2015	Korea: one of 3 within-switchers
Hong Kong; Macao (CEPA)	2003	excluded: entrepôt, sui generis

Entry-into-force years as coded in the World Bank Deep Trade Agreements database and TREND; see text. “Within-switchers” are destinations whose EP depth changes after initial entry: Laos (Bangkok Agreement 2002 → ASEAN 2005), Singapore (ASEAN 2005 → Singapore–China FTA 2009), and South Korea (Bangkok Agreement 2002 → 2015 FTA). Timor-Leste is coded as an ACFTA party in the source databases although it is not an ASEAN member; it accounts for 0.02% of observations and the main coefficients are unchanged to well beyond the fourth decimal if it is dropped (Online Appendix diagnostic).

Pakistan 2007, New Zealand 2008, Peru 2010, Costa Rica 2011, Australia 2015). Of the 236 destinations in the panel, 23 are ever treated and 213 are never treated over the sample window.

EP content is measured with two independent codings: the World Bank Deep Trade Agreements database (Hofmann et al., 2017), from which I build `WB_EP_Depth` (the count of environmental-law provisions in force with destination d at time t) and a non-environmental `TotalDepth` control; and TREND (Morin et al., 2018), from which I build `TREND_EP_Count` and sub-indices (green market access, enforcement, hard vs. soft provisions). Using two codings guards against idiosyncrasies of either; as shown below, they agree.

2.2 Customs data and product classification

The trade data are the universe of Chinese export customs transactions 2000–2015, aggregated to the firm–HS6–destination–year level: 49.2 million observations (45.8 million once Hong Kong and Macao are dropped), about 462,000 firms, roughly 5,000 HS6 products and 236 destinations. The outcome is the log of export value; the panel identifiers used for fixed effects (firm–product–destination, firm–destination–year, product–year) are precomputed. Export values enter in logs with no trimming or winsorization; Section 5’s inference battery, which is design-based rather than moment-based, is the primary guard against influential observations.

Table 2: Treatment-timing cohorts

Entry year	N destinations	Destinations
2002	5	Bangladesh, India, Korea, Laos, Sri Lanka
2005	10	Brunei, Cambodia, Timor-Leste, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam
2006	1	Chile
2007	1	Pakistan
2008	1	New Zealand
2010	1	Peru
2011	1	Costa Rica
2014	2	Iceland, Switzerland
2015	1	Australia
Total treated	23	
Never treated	213	

Excludes Hong Kong and Macao (2003 entry; entrepôt economies, excluded from the main specification, Section 2). South Korea’s 2015 upgrade from the shallow Bangkok Agreement to the deep bilateral FTA is a within-country depth increase, not a new entry, and so does not create a separate cohort here; it is the switch referenced in Table 1’s notes. Source: `New/Output/Diagnostics/B_treatment_entry.csv`.

Green products are the 248 HS6 codes of the OECD Combined List of Environmental Goods (CLEG) as compiled in Sauvage (2014) (verified match of 246/248 against the project’s source file, with the single discrepancy a granularity split of one HS heading into two sub-codes). Because the customs panel uses the HS1996 vintage while the OECD list is native HS2012, the list was translated to HS1996: 246 of the 248 codes have a unique 1:1 concordance and are translated, while the two that do not (871411 and 871419, the granularity split noted above) are retained at their original codes and flagged rather than dropped. Export-value continuity checks around the 2007 revision boundary find no suspicious break on any translated code.¹ *Dirty* products are the HS6 codes belonging to the classic pollution-intensive sectors of Mani & Wheeler (1998) and Low & Yeats (1992) — pulp and paper, industrial chemicals, petroleum refining, iron and steel, non-ferrous metals (plus cement in an extended list) — mapped to HS1996 through the official WITS/UNSD HS1996–ISIC Rev.3 correspondence: 1,139 codes. Seventeen codes appearing in both lists (e.g. insulation materials: produced by dirty industries, used for environmental purposes) are assigned to the green list, which is product-curated; the two categories are mutually ex-

¹ As a check on the classification itself rather than its translation, the main specification was re-estimated restricting green products to the 54 codes of the APEC Environmental Goods List (Sauvage, 2014), on which explicit multilateral political consensus exists and which form a complete subset of the OECD list. With the WB index the green coefficient flips sign to +0.0050 (s.e. 0.0127, $p = 0.69$), with the standard error roughly doubled, as expected from an 80% reduction in the green sample; with TREND it moves from +0.0018 to +0.0032 ($p = 0.13$), same sign and similar precision. Neither is distinguishable from zero: the null is not an artifact of borderline products in the broader list.

clusive. Green products account for 11.5% of firm-level observations and dirty products for 7.0%; the remaining “neutral” products form the comparison group.

Table 3: Descriptive statistics

Firm-level panel (excl. HK–MO)	
Observations (firm–HS6–dest–year)	45,781,211
Exporting firms	$\approx 462,000$
HS6 products	$\approx 5,000$
Destinations	236
of which treated (EP depth > 0)	23
Effective agreements	≈ 14
Share of observations: green products	11.5%
Share of observations: dirty products	7.0%
Share of observations under an EP agreement	20.3%
Collapsed panel (HS6–dest–year)	
Cells	3,773,498
green / dirty cells	8.4% / 14.0%
Zero-filled PPML grid (HS6–dest–year)	8,310,464
Hong Kong and Macao excluded (they account for 24.4% of treated observations and 50.1% of treated export value). Product classification as described in the text.	

2.3 Estimation samples and control-group subsamples

The analysis uses four estimation objects built from the same underlying panel, each answering a different question. The *full firm-level panel* (45.8 million firm–HS6–destination–year observations once Hong Kong and Macao are dropped) is the main object; its singleton-surviving estimation sample is characterized in Section 3. The *collapsed panel* aggregates it to HS6–destination–year cells (3.77 million; outcome: within-cell mean of log exports, regressions weighted by cell size) and serves both as a computational workhorse for the inference battery and as an internal cross-validation — its fixed-effects structure preserves the identifying logic but not the firm dimension, a difference Section 3 quantifies. The *zero-filled PPML grid* (8.3 million cells) adds the combinations with no recorded trade, opening the extensive margin. The *within-firm share panel* (13.3 million firm–destination–year observations) recasts the outcome as the green share of each firm’s export basket to a destination; it enters as descriptive evidence only, for the identification reason given in Section 5.

Because the triple-difference compares green and dirty products to *neutral* ones, the credibility of the comparison set can be probed directly. Four control-group subsamples, each tightening the comparison in the direction of a specific threat:

Table 4: Control-group subsamples

Subsample	Construction	Effective size	Threat probed
C-prod-HS4	neutral products in the same HS4 family as a green product (106 families; 20.5% of observations)	3.8M	unlike comparison products (technological similarity)
C-overlap	only HS6 products exported both to treated and to never-treated destinations ($\approx 98.5\%$ of products)	21.5M	extrapolation outside common support
CEM destinations	destinations matched by coarsened exact matching on year-2000 log GDP p.c., growth, and MFN tariff (16 treated + 40 controls)	13.7M	selection of partner countries
Deep vs. shallow	PTA partners only, split at the median of maximum EP depth (16 deep vs. 9 shallow in-sample)	5.3M	the “any agreement” contrast itself

Effective sizes after iterative singleton removal, as in Table 7. C-prod-HS4 restricts the product margin and is read alongside C-overlap, never alone, because a within-firm reallocation spillover would contaminate it: multi-product firms concentrate on core products when trade costs fall (Eckel & Neary, 2010), so a neutral product in the same HS4 family, sold by the same firm, can be pushed down by the very reallocation the treatment induces — which biases the green-versus-neutral contrast away from zero rather than toward it. C-overlap is immune by construction, which is why the two are reported together. The deep-vs-shallow design, in the spirit of the within-treated contrast of Abman et al. (2024), drops untreated destinations entirely, so the identifying comparison is between deeper and shallower environmental chapters among agreement partners; its limitation is the small number of shallow clusters (9), which the few-cluster inference battery of Section 3 is designed to handle.

The reading rule, developed in Section 4: a genuine composition effect should survive every tightening, while an artifact of the comparison set should move as the comparison set moves. Stability across these subsamples is therefore not a formality but part of the result.

3 Empirical strategy

3.1 Why a level effect is not identifiable

The natural first specification — regressing log exports on EP depth with fixed effects — is uninformative here, and I document this as a diagnostic rather than as a result. EP depth varies at the destination–year level and switches on with the agreement; with ~ 14 agreements and near-zero within-destination variation, “environmental depth” and “having a (deep) agreement” cannot be told apart. The saturation ladder makes the point empirically: moving from sparse (firm–product–destination + year) to saturated (adding product–time and pair-level interactive effects) structures, the EP-depth coefficient falls monotonically from small positive and starred to an exact, precisely estimated zero. The pattern — significance that lives only in under-saturated, finely-clustered specifications — is the classic signature of selection combined with understated standard errors (Bertrand et al., 2004; Goodman-Bacon, 2021). One consequence is worth stating plainly: this design cannot supply the kind of first-stage “bite” evidence — a demonstration that signing an EP-laden PTA moved some intermediate outcome, such as aggregate trade volumes, before composition is examined — that would normally open a difference-in-differences case, because that level effect is exactly what is confounded with the agreement. The saturation ladder is offered as the closest available substitute: it shows that whatever the agreement did at the level of trade, EP depth specifically is not separably identified there, which is why the paper’s evidence is built entirely on the composition margin instead.

3.2 The triple-difference on composition

The target parameter is an average treatment effect on the treated: the effect of one additional unit of environmental depth on the gap between green (or dirty) and neutral log export value, within firm–destination–year cells that experience treatment: in potential-outcomes terms, $\beta_1 = E[(y_{fgpdt}(1) - y_{fgpdt}(0)) - (y_{fnpdt}(1) - y_{fnpdt}(0)) \mid d, t \text{ treated}]$, where $y(1)$ and $y(0)$ denote log exports of green (g) versus neutral (n) products under, respectively, deeper and unchanged environmental content, holding the agreement itself fixed. This is a per-cell (per firm–destination–year), not a per-firm, average. The collapsed-panel implementation aggregates the firm dimension with cell-count weights, so that destinations or years with more

underlying transactions contribute proportionally more evidence; the weights are the cell counts of underlying transactions, so the weighted cell-level regression is algebraically identical to the unweighted micro-level regression under the same fixed effects (Section 3 verifies this to seven significant figures). Weighting is thus not a modelling choice but the condition for that equivalence, and no weighting is by the outcome itself. The main specification is

$$\ln x_{f p d t} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + \theta_{f p d} + \theta_{f d t} + \theta_{p t} + \varepsilon_{f p d t}, \quad (1)$$

where f indexes firms, p HS6 products, d destinations, t years; g_p and b_p are the green and dirty indicators, EP_{dt} is environmental depth and TD_{dt} non-environmental agreement depth. The firm–destination–year effects $\theta_{f d t}$ are the key: they absorb everything that varies at the level of the firm’s relationship with a market in a year — including the agreement itself, market size, demand shocks, and selection into agreements. Identification comes from comparing green (and dirty) products to neutral ones *within* the same firm–destination–year, as EP content varies across destinations and over time; $\theta_{p t}$ absorbs global product-level shocks (e.g. the world solar boom) and $\theta_{f p d}$ the level of specialization. Mechanically, $\theta_{f d t}$ places a separate intercept on every firm–destination–year cell: any variable constant within that cell — the agreement dummy, its overall depth, destination demand, and the non-random selection of destinations into agreements — is perfectly collinear with those intercepts and drops out. Selection into agreements can therefore contaminate β_1 only if it operates *differentially* on green versus neutral products within the same firm–destination–year, the residual threat that the destination-trend exercise of Section 5 targets. The TD interactions separate environmental content from general agreement depth, the same role the DESTA index (Dür et al., 2014) plays in Brandi et al. (2020); because **TotalDepth** counts all non-environmental provisions rather than isolating green-relevant ones (e.g. subsidy disciplines, technical-standard harmonization), it is an imperfect proxy for the depth of agreement content that is genuinely orthogonal to the environmental chapter. An imperfectly measured control does not attenuate the coefficient on the variable of interest in a signable direction: it under-controls, and β_1 absorbs some of the effect of general depth with no guarantee about the sign. Rather than assert a direction, Section 5 reports the coefficient under four different depth controls — including none at all — and shows that the point estimate moves within a band narrower than one standard error (Table 9). That is evidence about the direction, where an appeal to attenuation would only have been an assumption. The correlation is substantial and should be stated plainly: across the 223 treated destination–years in the estimation sample, WB EP depth and non-environmental TotalDepth correlate at 0.91 in the raw data and 0.96 after destination and year demeaning (the variation the design actually uses); the raw-data VIF

of 5.8 is somewhat higher than the maximum of 4.6 that Brandi et al. (2020) report. The $EP \times \text{green}$ coefficient is therefore identified from the modest share of environmental-depth variation that is orthogonal to general agreement depth — a limitation that the clustered standard errors already price in (it is one reason the WB confidence intervals are wider than the TREND ones: the TREND count is far less collinear with TotalDepth, at 0.50 raw and 0.85 within). That two codings with such different degrees of overlap with general depth return the same null is itself reassuring: a result driven by the inability to separate environmental from general content would be expected to differ across them. Standard errors are clustered by destination, the level of treatment variation (Abadie et al., 2023).

One qualification is due, and it concerns the *estimator*, not the target; it matters for how β_1 should be read. Environmental depth is a *continuous* dose that switches on at *staggered* dates, and under that combination the two-way fixed-effects coefficient is not in general the ATT defined above: it is a weighted average of dose-specific and cohort-specific effects whose weights need not be convex, and comparisons across dose levels are additionally exposed to selection into dose that parallel trends alone does not rule out (Callaway et al., 2024; Goodman-Bacon, 2021; de Chaisemartin & D’Haultfœuille, 2020). This paper does not implement a continuous-dose robust estimator; the Sun–Abraham event study of Appendix A addresses only the staggered-timing half of the problem, and it does so on a binarized treatment. Two considerations limit how much this costs here. Estimating a null makes the weighting problem less consequential than it would be for a non-zero estimate — non-convex weights distort the aggregation of heterogeneous effects, but a weighted average of effects that are individually near zero stays near zero unless the underlying effects are large and offsetting, which the sub-index and leave-one-out evidence gives no sign of. And the permutation test of Section 3 is agnostic about the weighting entirely: it asks whether the observed coefficient, whatever weighted average it represents, is unusual relative to the same statistic computed under randomly reassigned EP profiles. β_1 is therefore best read as a weighted average of composition responses across doses and cohorts, not as a single structural ATT, and a continuous-dose estimator in the spirit of Callaway et al. (2024) remains the natural next step for this design.

A further identification concern works in a conservative direction. If deeper environmental chapters are more likely to be signed with destinations where consumer or regulatory demand for green goods is *already* rising — a plausible selection story — then EP_{dt} is positively correlated with an omitted destination-specific green-demand trend, which would bias β_1 upward, toward finding a spurious positive effect on green exports. Because the estimated β_1 is a precise zero rather than positive, this channel implies the true effect (if negative or zero) is, if anything, understated by the null: the null is a conservative, not a lenient, reading of

the data.

Two complementary implementations address the computational burden of three high-dimensional fixed effects on 46 million rows. The full panel is estimated with iterative singleton removal (Correia, 2017), which eliminates 24.3 million uninformative observations. The surviving sample is not a selected sliver: it retains 47% of observations but 70% of aggregate export value (the removed observations are predominantly one-off, small flows), and the product composition is essentially unchanged (green products are 11.5% of observations before removal and 11.9% after; dirty products 7.0% and 6.4%). Within the estimation sample, the composition contrast is identified in the treated firm–destination–year cells that contain both a green (or dirty) and a neutral product: 26% of treated cells for the green margin and 12% for the dirty margin, with a median of one green (dirty) product per identifying cell and three products per treated cell overall — a reminder that the estimates aggregate many small within-firm contrasts rather than a few large ones. A collapsed panel at the HS6–destination–year level (3.77 million cells; outcome: within-cell mean of log exports; regressions weighted by cell size; fixed effects $pd + dt + pt$), in the gravity-equation tradition of Head & Mayer (2014), reproduces the design at a fraction of the cost, with dt playing the role that fdt plays in the full panel: it absorbs the agreement.

What it does not reproduce is the firm dimension, and the two panels should be compared with that in mind rather than treated as interchangeable. Re-estimating the full panel with $pd + dt + pt$ in place of $fpd + fdt + pt$ recovers the collapsed coefficients to seven significant figures (-0.0045685 vs. -0.0045685 on green; -0.011873 vs. -0.011873 on dirty), which establishes that the weighted collapse is algebraically the micro-regression under those fixed effects and that the two pipelines agree exactly where they are asked the same question. It also isolates what the firm effects do. On the green margin the two structures agree substantively (-0.0046 collapsed, -0.0023 full panel; both indistinguishable from zero). On the dirty margin they differ by a factor of 2.7 (-0.0119 vs. -0.0044), and the gap is a result rather than a discrepancy: the collapsed specification compares products *across* the firms serving a destination–year, so any shift in which firms export dirty goods enters the coefficient; fdt removes that channel and leaves only reallocation within the firm. Roughly three-fifths of the collapsed dirty coefficient is therefore between-firm composition, not within-firm response — consistent with Section 4.4’s reading, and a reason the full panel is the reference specification.

3.3 Inference with few treated clusters

There are 225 destination clusters in the full-panel reference specification (236 in the collapsed panel) but only 23 treated ones, and effectively 14 agreements. Asymptotic cluster-robust inference is unreliable in this regime, so every headline estimate is subjected to three

tests: (i) destination-clustered asymptotics; (ii) wild cluster bootstrap with 9,999 replications (Cameron et al., 2008; Fischer & Roodman, 2021; Roodman et al., 2019) — computed on Frisch–Waugh-residualized data (fixed effects demeaned once, not re-estimated at each bootstrap draw). This carries two approximations, both in the collapsed-panel implementation only. It treats *pt* as if it were nested within the destination cluster, which it is not, while *pd* and *dt* are. And the small-sample corrections see the four regressors of the residualized equation rather than the four plus the absorbed fixed effects, so the degrees-of-freedom adjustment is not exact. Demeaned coefficients match the unrestricted **fixest** estimates exactly, so the point estimates are unaffected. Neither approximation applies to the full-panel bootstrap, which runs natively after **reghdfe** and is the version quoted for the headline intervals in Section 4. With only 23 *treated* clusters among 236, the more demanding regime of MacKinnon & Webb (2017) and Conley & Taber (2011) applies: standard wild bootstrap asymptotics assume the number of treated clusters, not just clusters overall, grows large, which is why (iii) permutation inference (Fisher, 1935) is run as an independent check rather than a formality. Entire EP profiles (depth and timing) are randomly reassigned 1,000 times across treated destinations, asking whether the observed coefficient stands out among placebos — the sharpest answer to “is it really the environmental content, or would any labelling of these destinations produce this number?”. Three features of the test should be stated rather than left implicit. The null is the restricted one — which treated destination holds which profile — not treated versus untreated, since never-treated destinations are never given a profile. Depth and timing are permuted jointly, so the test asks about the whole treatment assignment and cannot separate environmental content from the date of entry. And the effective support is far smaller than 1,000: the eleven ASEAN destinations carry identical profiles, so permuting them among themselves changes nothing, leaving roughly nine distinct profiles. The permutation distribution is correspondingly granular, and the resolution of the resulting *p*-values is bounded by that, not by the number of draws. The dynamic specification is an event study of the green (dirty) vs. neutral differential around entry into force, with never-treated destinations as controls, endpoint bins declared, and a Sun–Abraham version (Sun & Abraham, 2021) run on the destination-level composition gap to address staggered-timing cohort heterogeneity (Goodman-Bacon, 2021; de Chaisemartin & D’Haultfœuille, 2020; Callaway & Sant’Anna, 2021).

4 Main results

4.1 The green margin: a stable, precise null

Table 6 reports the main estimates. In the full firm-level panel the $\text{EP} \times \text{green}$ coefficient is -0.0022 (s.e. 0.0039) with the WB index and -0.0001 (s.e. 0.0010) with TREND; the joint F-test on all four composition interactions is $p = 0.31$ and $p = 0.71$ respectively. The collapsed panel returns -0.0046 (WB): the wild cluster bootstrap p-value is 0.65, and permutation inference on the exact estimating equation — 1,000 reassignments of entire EP profiles across treated destinations — gives $p = 0.61$: the observed coefficient is well within the bulk of the placebo distribution.

To gauge magnitudes, it helps to benchmark the confidence intervals against the literature’s reference estimate (Table 5). Brandi et al. (2020) find that one *liberal* environmental provision raises the green share of developing-country exports by 0.4 percentage points, about +17% of the mean green share — roughly +0.16 log points of green relative to non-green exports per provision in the metric of equation (1). Section 3 argued that asymptotic inference is unreliable with 23 treated clusters; that argument applies to interval estimates as much as to p-values, so the bounds quoted here are wild cluster bootstrap intervals. The full-panel 95% bootstrap confidence interval for $\text{EP} \times \text{green}$ is $[-0.0353, +0.0355]$ per WB provision (collapsed panel: $[-0.0182, +0.0317]$). The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly six times narrower, and reporting precision on that basis would claim exactly the kind of certainty this paper’s own inference section declares unwarranted. On the bootstrap bound, the upper limit is about one quarter of the Brandi-equivalent effect: the design rules out their point estimate and anything above roughly a quarter of it, and is uninformative below that threshold. This is a weaker statement than the asymptotic interval would support, and it is the one the design can sustain. On the dirty margin their trade-restrictive provisions reduce dirty exports by roughly 5% of the mean per provision (≈ -0.05 log points); even taking our asymptotic full-panel point estimate at face value (-0.0044 per provision), the response is an order of magnitude smaller. Both comparisons carry the same caveat in both directions: Brandi et al.’s effects are identified from explicitly liberal or trade-restrictive clauses, which are almost entirely absent from Chinese agreements of this period (Section 5) — which is precisely the interpretation the paper advances. In per-standard-deviation terms, a one-SD increase in WB EP depth in the estimating sample (≈ 2.383 provisions, weighted by cell size and inclusive of never-treated destination-years at $\text{EP}=0$) moves green relative to neutral exports by at most -8.4% to $+8.5\%$ at the 95% bootstrap bounds (-2.4% to $+1.3\%$ on asymptotic bounds).

Table 5: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	-0.0513	-0.0044 (point est.)	–	1/12

Table 6: Triple-difference estimates: EP content and export composition

	WB EP depth		TREND EP count	
	× green	× dirty	× green	× dirty
Full panel (reghdfe)	−0.0023 (0.0039)	−0.0044 (0.0022)	−0.0001 (0.0010)	−0.0009 (0.0006)
asymptotic p	0.57	0.052	0.92	0.15
wild cluster bootstrap p	0.69	0.18	0.93	0.18
bootstrap 95% CI	[−0.035, +0.036]	[−0.043, +0.011]	[−0.0037, +0.0025]	[−0.0027, +0.0007]
Collapsed panel	−0.0046 (0.0070)	−0.0119*** (0.0030)	+0.0018 (0.0018)	+0.0004 (0.0016)
asymptotic p	0.51	<0.001	0.32	0.83
wild cluster bootstrap p	0.65	0.07	0.39	0.85
bootstrap 95% CI	[−0.018, +0.032]	[−0.018, +0.002]	[−0.0018, +0.0071]	[−0.0032, +0.0054]
permutation p	0.61	0.23	0.18	0.85

Full panel: 21,519,511 obs. after iterative singleton removal, FE $f_{pd} + f_{dt} + p_t$, 225 destination clusters; joint F on the four interactions: $p = 0.31$ (WB), $p = 0.71$ (TREND). Collapsed panel: 3,681,023 cells (3,773,498 before fixed-effect singleton removal), FE $p_d + d_t + p_t$, weighted, 236 clusters. All specifications include TotalDepth×green/dirty controls. Wild cluster bootstrap: B=9,999 (Cameron et al., 2008). Permutation: 1,000 reassignments of entire EP profiles across treated destinations, re-estimated on the exact collapsed specification of this table; p-values are the share of placebo assignments producing a larger absolute coefficient. A coarser variant on a destination–year–product-type aggregate gives $p = 0.40$ for green and reverses the dirty sign (+0.005, $p = 0.49$): both versions leave the dirty coefficient indistinguishable from its placebo distribution; Section 4.4 discusses the sign reversal under aggregation. Bootstrap intervals, not asymptotic ones, are the basis for every magnitude claim in the text: with 23 treated clusters the asymptotic intervals are roughly six times too narrow (Section 3). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (asymptotic only; does not survive wild cluster bootstrap or permutation inference — see text).

4.2 Stability across control groups

Table 7 re-estimates equation (1) on the four control-group subsamples described in Section 2.3 (Table 4), each of which tightens the comparison group in the direction of a specific threat: C-prod-HS4 (the product-family restriction), the CEM destination match (Iacus et al., 2012), the deep-vs-shallow within-treated contrast (Abman et al., 2024), and common support. Together with the two baseline designs (full and collapsed panel) and three further full-panel sample variations (adding controls, excluding ASEAN, including Hong Kong–Macao), the table reports nine rows in total. The green coefficient barely moves — between -0.0009 and -0.0046 — and is never significant. The stability is the point: a comparison-group artifact would move with the comparison group.

Table 7: Stability of the $EP \times \text{green}$ coefficient across estimation designs (WB index)

Design	Effective sample	$EP \times \text{green}$	p
Full panel (firm level)	21.5M	-0.0023	0.57
Collapsed panel	3.7M cells	-0.0046	0.51
C-prod-HS4 (within green HS4 families)	3.8M	-0.0009	0.86
CEM-matched destinations	13.7M	-0.0023	0.54
Deep vs. shallow (PTA partners only)	5.3M	-0.0022	0.53
Common support (C-overlap)	21.5M	-0.0022	0.57
Full panel, with controls	19.3M	-0.0002	0.94
Full panel, excluding ASEAN	19.2M	-0.0025	0.48
Full panel, including HK–Macao	23.6M	-0.0009	0.80

Each row estimates equation (1) on the indicated subsample; destination-clustered s.e. CEM match on log GDP p.c., growth, MFN tariff (year 2000). Deep/shallow split at the median of maximum EP depth (16 deep vs. 9 shallow countries in the estimation sample, which excludes Hong Kong and Macao). The common-support restriction is reported for completeness rather than as an independent test: 98.5% of HS6 products are exported both to treated and to never-treated destinations, so it removes 314 of 21.5 million observations and reproduces the full panel almost exactly. Its value is in showing that extrapolation outside common support is not an issue here, not in providing a distinct estimate. TREND-index analogues (where estimated) give the same picture: full panel -0.0001 ($p = 0.91$), common support -0.0001 ($p = 0.91$), deep vs. shallow -0.0004 ($p = 0.72$). Controls: MFN tariff, market concentration (HHI), antidumping indicator.

4.3 Dynamics

Figure 1 plots the event study. Differential pre-trends are flat for both green and dirty products, and nothing happens at entry into force. This is a falsification check, not a proof: parallel trends is a statement about the unobserved counterfactual path of treated destinations after treatment, which no pre-period test can directly verify; a flat pre-trend rules out

the most detectable violations (an already-diverging composition gap) without establishing that the assumption holds exactly thereafter. The late negative drift for green products (bin $\geq +5$) is identified only by early cohorts (ASEAN 2005, Chile 2006, Pakistan 2007, New Zealand 2008) and does not survive the Sun–Abraham estimator applied to the destination-level composition gap: the aggregated ATT is -0.044 ($p = 0.24$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap — consistent with cohort heterogeneity rather than a treatment effect (Sun & Abraham, 2021). That estimator answers a narrower question than equation (1) and should not be read as confirming it: it binarizes treatment (EP present versus absent, discarding depth), carries no depth control, and runs on a destination–year composition gap built by collapsing across products, so it has no product–year fixed effects. It is a diagnostic for staggered timing and cohort heterogeneity — which is what it is used for here — not an independent replication of the static specification. The full Sun–Abraham event study is reported in Appendix A; one distant lead (the dirty gap at $t = -6$) is individually significant, and the appendix shows through a cohort decomposition that it reflects the idiosyncratic deviation of a single late-cohort destination rather than a common pre-trend.

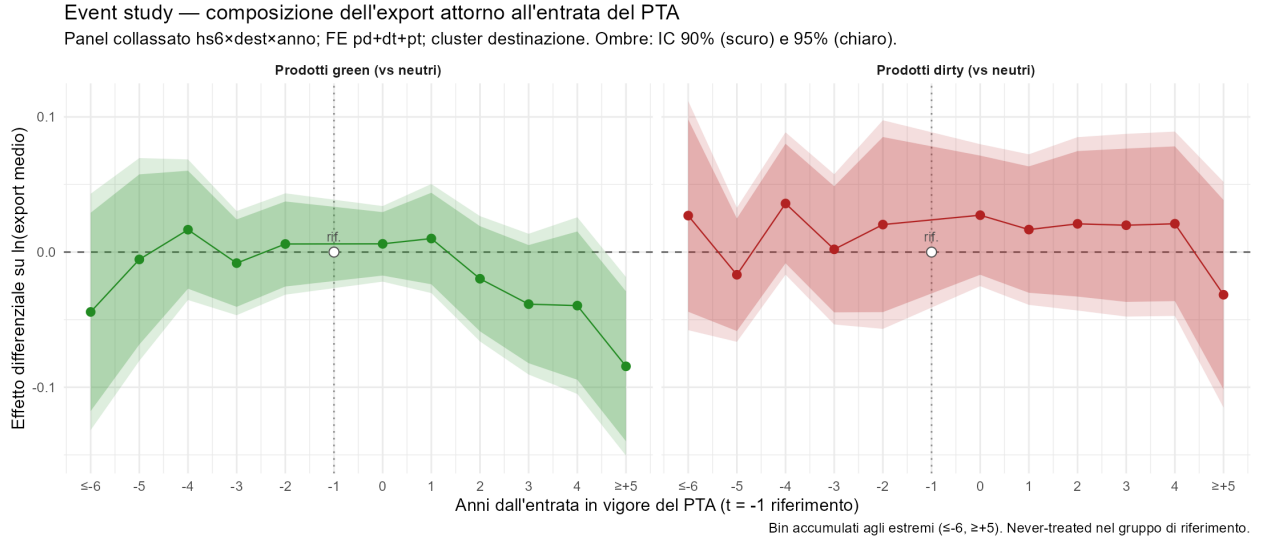


Figure 1: Event study: green and dirty vs. neutral products around PTA entry into force. Collapsed panel; FE $pd + dt + pt$; destination-clustered 90% (dark) and 95% (light) bands; $t = -1$ is the reference period; endpoint bins accumulate.

4.4 The dirty margin: anatomy of a false positive

The strongly significant $WB \times \text{dirty}$ coefficient in the collapsed panel (-0.0119 , asymptotic $p < 0.001$) deserves its own subsection, because it is a textbook illustration of why few-cluster inference matters: every robust check the coefficient is put through returns it to

the null. The wild cluster bootstrap raises its p-value to 0.07, just above the conventional 5% threshold. Permutation inference on the exact estimating equation agrees: reshuffling entire EP profiles across treated destinations, 23.5% of placebo draws produce a coefficient as large in absolute value ($p = 0.23$), placing the observed estimate squarely inside the placebo distribution. A coarser permutation on a destination–year–product-type aggregate reverses the sign entirely ($+0.005$, $p = 0.49$): aggregating to one observation per destination–year–product-type collapses the within-cell heterogeneity across firms and product varieties that the disaggregated coefficient exploits, so the two permutation tests are not answering quite the same question — but they reach the same verdict, and the sign instability under aggregation is further evidence of fragility. A leave-one-out exercise across the 23 treated destinations points the same way: the coefficient is stable (≈ -0.012) except when Australia is dropped, in which case it falls to -0.0103 with $p = 0.24$. South Korea, the destination that drove the equivalent result in an earlier version of this analysis, is no longer pivotal on its own: dropping it leaves the coefficient at -0.0097 with $p = 0.09$. The TREND index never shows the effect (bootstrap $p = 0.86$). In the full firm-level panel the coefficient is -0.0044 with an asymptotic p of 0.052 — just above conventional significance, the same order of magnitude as in the CEM subsample — and the wild cluster bootstrap on that same panel moves it to $p = 0.18$, with a 95% interval of $[-0.043, +0.011]$ that comfortably contains zero. The marginal asymptotic signal is thus erased by robust inference on both panels independently, not only on the collapsed one. The honest reading is that the dirty margin is a false positive of exactly the kind few-cluster designs manufacture: a coefficient that is overwhelming under asymptotic standard errors ($p < 0.001$) and that no form of robust inference sustains. It belongs in the paper as a descriptive pattern and a methodological warning, not as an identified effect. Section 5 shows this is compounded by the near-total absence of trade-restrictive content in Chinese agreements’ dirty-relevant chapters, the same structural limitation that governs the green margin.

5 Robustness and extensions

5.1 Provision bundling and the limits of heterogeneity analysis

A natural objection is that the aggregate index dilutes provisions with a trade mechanism (green-goods liberalization; binding standards) with cooperation language that has none. The objection is correct in principle — and unanswerable in this setting, which is itself a finding. The imbalance can be quantified directly, and it is large in both codings. Summing WB_EP_Depth across the 25 treated destinations gives 150 provisions checked in

total; only 5.3% of them fall in `GreenLiberalization` or `StandardsNonRegression`, the two sub-indices with a direct trade mechanism. TREND tells the same story on an independently coded, far more granular scheme: of 437 provisions checked across the same destinations, `GreenMarketAccess` accounts for 0.92%. At the agreement level, TREND’s own binding/non-binding distinction (`X5.01.01 Binding obligations`) is coded present in exactly one of the 14–15 Chinese agreements in the database (China–Korea, 2015); every other agreement scores zero on that specific clause. Table 8 summarizes both codings side by side. What follows is a mechanism argument *for the null*, not evidence of a positive causal channel: it explains why no effect is detected, rather than tracing how an effect would operate if one existed. Across the 223 treated country–year observations in the estimation sample, the two WB sub-indices with a direct mechanism (`GreenLiberalization` and `StandardsNonRegression`) are *perfectly* collinear ($\rho = 1.000$) — and the reason is stark: both are non-zero in only three country–years (Korea from 2015; Switzerland from 2014), always in the same 1:3 proportion. Mechanism-bearing provisions effectively exist in two Chinese agreements of the period. Their interactions load identically in the regression (identical t-statistics; coefficients in exact 3:1 ratio). TREND sub-indices correlate between 0.5 and 0.9 with each other and with the total count. With ~ 14 agreements, provision-type heterogeneity is not separately identifiable — in sharp contrast with Brandi et al. (2020), whose 680-agreement panel allows it (they report a maximum VIF of 4.6 across their provision-type variables; here, for the two mechanism-bearing sub-indices, it is unbounded). One reading rule applies to everything that follows: each sub-index is estimated *on its own*, replacing the aggregate index one at a time, not entered jointly. The coefficients are therefore not partial effects holding other clause types constant — each absorbs whatever variation in the correlated sub-indices moves with it. Entering them jointly is precisely what the collinearity documented above rules out, so this is the only feasible reading, but it means the table is a set of separate comparisons and not a decomposition of the aggregate index into additive parts. What can be said: the mechanism-bearing bundle shows the same pattern as the aggregate index — green never responds (green-liberalization $\times$ green: $p = 0.90$; TREND green-market-access $\times$ green: $p = 0.38$), while the dirty interaction is marginally negative on asymptotic p-values ($p = 0.04$ – 0.07), the same asymptotics-only signal as the aggregate — which Section 4.4 shows does not survive robust inference. The two placebo components with no trade mechanism behave differently, and the difference is worth stating plainly rather than averaging away. TREND soft provisions show nothing on either margin ($\times$ green: $p = 0.27$; $\times$ dirty: $p = 0.86$), as the selection logic would predict. Regulatory-space clauses do not: they carry $+0.024$ on green and $+0.023$ on dirty, and both survive the wild cluster bootstrap ($p = 0.046$ and $p = 0.022$ respectively, $B=9,999$). This is the only place in the

paper where a component with no trade mechanism registers a signal that robust inference sustains, and it is reported here as a caveat, not resolved. Two features bound what can be read into it. First, the two margins move together: the green-dirty differential is +0.0017 with $p = 0.80$, so the estimate describes green and dirty exports rising by the same amount against the neutral baseline — a pattern that is not the signature of a shift in composition toward cleaner goods, which is the object this paper measures. Second, and more fundamentally, this sub-index is not a clean placebo to begin with. Regulatory-space clauses account for 71.5% of the TREND count across treated destination-years and correlate 0.90 with the TotalDepth control; in this specification the depth control itself flips to significantly negative ($\times$ green: -0.0011 , $p = 0.006$). Two regressors correlated at 0.90 splitting into a large positive and a large negative coefficient is this section’s bundling problem applied to the placebo, not an independent channel measured cleanly. The defensible conclusion is the weaker one: this design cannot separate a regulatory-space effect from general agreement depth, which means it cannot use that sub-index as a clean falsification test either. Enforcement provisions — dispute-settlement mechanisms attached to the environmental chapter, which condition whether any of the above is binding rather than providing a trade mechanism themselves — are likewise null on both margins and both codings (WB **EnforcementDSM** $\times$ green: $p = 0.91$; $\times$ dirty: $p = 0.90$; TREND **EnforcementDSM** $\times$ green: $p = 0.78$; $\times$ dirty: $p = 0.71$).

Table 8: Trade-mechanism content as a share of all checked provisions

	Total checked	Trade-mechanism	Share
WB (GreenLiberalization + StandardsNonRegression)	150	8	5.3%
TREND (GreenMarketAccess)	437	4	0.92%
	Agreements	With binding clause	
TREND (X5.01.01 Binding obligations)	14–15	1 (China–Korea, 2015)	6.7–7.1%

“Total checked” sums the relevant depth index (WB_EP_Depth or TREND_EP_Count) across the 25 treated destinations, using each destination’s maximum (post-entry) value. “Trade-mechanism” sums the corresponding sub-index over the same destinations. The binding-clause row is agreement-level, not destination-level, which is why the agreement count differs from 25 (see Table 2’s notes on ASEAN pooling and within-switchers). Source: `Data/Merged/Merged_TREND_WB_Indices_Only.csv` and `Data/TREND/TREND_China_2000_2015.csv`.

This bundling result also reconciles the null with Brandi et al. (2020)’s positive finding, even though both papers draw on the same underlying TREND/World Bank provision coding. Their effect is not a generic “environmental provisions matter” result: it is driven specifically by the subset of agreements containing explicitly trade-liberalizing or trade-restrictive environmental clauses, identified separately from cooperation-only language because their 680-agreement panel has enough cross-agreement variation in provision *type* to support that decomposition (maximum VIF 4.6 across provision-type variables). The Chinese

PTAs of 2000–2015 are, on this same coding scheme, overwhelmingly the cooperation-only type: mechanism-bearing content — the exact category driving Brandi et al.’s result — appears in only 2 of the 14 agreements, and is perfectly collinear across its two components in-sample. With that little variation in provision type, this design cannot separately identify a mechanism-bearing effect even if one exists, and a null on the aggregate index is what Brandi et al.’s own heterogeneity would predict for a sample built almost entirely from cooperation-only chapters. The two results are therefore complementary rather than contradictory: theirs shows that mechanism-bearing provisions move trade when agreements contain them in sufficient number and variety; this paper shows that China’s agreements, over this period, did not contain them.

A second, compounding difference is the level of aggregation at which the two designs operate. Brandi et al. (2020) work with UN Comtrade flows aggregated to the exporter–importer–year level, with dirty/green export shares as the outcome; a composition effect there can arise simply because the set of firms or products exporting changes, without any single firm reallocating its own mix. The data used here are firm–product(HS6)–destination–year transactions, and the preferred specification saturates firm $\times$ destination $\times$ year fixed effects, so the EP coefficients are identified from within-firm reallocation across products at a given destination and year — a strictly more demanding test of the same composition hypothesis, and one on which a country-pair-level aggregation could in principle mask heterogeneity that this design is built to detect.

5.2 Extensive margin

Log-linear estimates condition on positive flows. If EPs created green trade at the extensive margin — combinations with no recorded trade before — the intensive estimates would miss it (Santos Silva & Tenreyro, 2006; Larch et al., 2025). The test available here operates at the product–destination level, not the firm level: on a zero-filled HS6–destination–year grid (8.3 million cells), it detects green products newly exported to a destination, but not a new firm beginning to export a product that other Chinese firms already ship there — which is the more common form of entry. The extensive margin is therefore probed, not exhausted. PPML estimates of equation (1)’s aggregated analogue confirm the null: EP $\times$ green is +0.0015 ($p = 0.74$) with the WB index and +0.0001 ($p = 0.95$) with TREND; EP $\times$ dirty is −0.030 ($p = 0.06$) and +0.003 ($p = 0.55$) respectively. There is no green trade creation at the extensive margin, and the dirty-margin coefficient here is, if anything, marginally stronger than in the intensive-margin specifications — consistent with the false-positive reading of Section 4.4: it does not clear the conventional 5% threshold on this grid either.

5.3 Within-firm reallocation: a descriptive complement

The margin that aggregate data cannot see: does the green *share* of a multiproduct firm’s basket toward a destination respond to EP depth? On the firm–destination–year panel of green shares (13.3 million observations; FE: firm–destination and year; destination-clustered s.e.), the coefficient on WB EP depth is -0.0001 ($p = 0.50$) and on the TREND count -0.00006 ($p = 0.043$, asymptotic). Both are economically negligible: a one-standard-deviation increase in the TREND count moves the within-firm green share by three hundredths of a percentage point, with the “wrong” sign, and the marginal asymptotic significance is subject to the same few-clusters caveat as everything else in this setting.

This module is reported as a descriptive complement, not as identifying evidence, and the reason is the one given in Section 3. The specification regresses the green share on the *level* of EP depth with firm–destination and year fixed effects; it does not include the firm–destination–year absorption that the triple difference relies on, and it therefore inherits precisely the confound that Section 3 argues is fatal to any level specification here — EP depth switching on with the agreement, and “environmental depth” being indistinguishable from “having a deep agreement” across ~ 14 agreements. A null recovered from a design that cannot identify an effect is not evidence that no effect exists; it is uninformative in both directions, and the marginally significant TREND coefficient illustrates the point rather than undercutting it. What the module does establish is descriptive and worth having: the within-firm green share is flat with respect to EP depth in the raw conditional means, which is at least consistent with the identified composition null and would have been a puzzle had it pointed the other way. It is not a firm-level counterpart to the within-plant evidence of Cherniwchan (2017), whose NAFTA tariff variation identifies a level effect that this setting cannot; establishing that comparison would require re-specifying the share equation as a triple difference with firm–destination–year effects, which the current design leaves for future work.

5.4 How much the depth control decides

Environmental depth and general agreement depth move together at 0.96 within destination and year, and any regression containing both must split their shared variation between them. How it splits depends on which depth measure is used, so a single coefficient understates the real uncertainty. Table 9 therefore reports the green coefficient under four depth controls rather than one: none at all, the aggregate WB `TotalDepth`, a targeted version restricted to the 14 non-environmental areas that actually correlate with EP content, and the externally-coded DESTA index (Dür et al., 2014).

Table 9: EP×green under alternative depth controls (collapsed panel, WB index)

Depth control	EP×green	95% CI
None	−0.0057	[−0.0118, +0.0003]
TotalDepth aggregate (17 areas) — <i>main specification</i>	−0.0046	[−0.0182, +0.0091]
TotalDepth targeted (14 areas)	−0.0033	[−0.0188, +0.0123]
DESTA index (independent source)	−0.0043	[−0.0129, +0.0042]

Collapsed panel, fixed effects $pd + dt + pt$, weighted by cell size, destination-clustered. Asymptotic intervals. The targeted control drops the three WB areas that do not correlate with EP depth (labour-market regulation, which has zero variance in Chinese agreements; visa and asylum; subsidies). DESTA covers seven policy areas coded independently of the World Bank instrument and excludes environmental provisions by construction, which is why it lowers the VIF from 5.7 to 1.9.

The point estimate moves within a band of 0.0024 log points — narrower than a single standard error of the main specification — stays negative throughout, and every interval contains zero. Dropping the depth control altogether, which is the specification most exposed to confounding by general agreement depth, moves the coefficient away from zero rather than toward it. The choice of depth control therefore does not decide the green result: the collinearity inflates the standard errors, as Section 3 says, but it does not put the point estimate at the mercy of an arbitrary modelling choice. Reporting the range is more informative than defending any one row of it.

5.5 Environmental share of agreement content

The main specification measures the marginal effect of one more environmental provision. A different question — the one Abman et al. (2024) pose — is whether, *given* that an agreement exists, a more environmentally-weighted agreement moves composition. That contrast requires restricting to PTA partners and replacing the level of EP depth with its share of total depth, EP_{dt}/TD_{dt} . On the collapsed panel restricted to the 25 partner destinations (534,846 cells; the share takes 12 distinct values in $[0.012, 0.068]$), the green interaction is −2.25 (s.e. 1.15, $p = 0.06$) and the dirty interaction −1.60 (s.e. 1.55, $p = 0.32$). Two cautions apply, and together they keep this from being read as a result. The estimand changes: this is the effect of the environmental *composition* of an agreement, not of an additional clause, so the coefficients are not on a scale comparable with the rest of the paper. And the share varies far less than the level does (coefficient of variation 0.19 against 0.62 on the same destinations), so the standard errors are correspondingly wide and the green coefficient clears neither conventional threshold despite its size. It is reported because the contrast is the one the closest antecedent in the literature uses, and because a marginally negative green coefficient is the opposite of what a green trade-creation story predicts — not because this

design identifies it well.

5.6 Sample robustness

Table 10 collects the full-panel sample variations. Adding controls (MFN tariff, market concentration, antidumping exposure) moves the green coefficient to an exact zero (-0.0002 , $p = 0.94$). Excluding the eleven ASEAN destinations — the block that dominates the treated count, and whose single 2005 accession date means the pooled coefficient could in principle be answering an ASEAN-specific rather than a general question — leaves it at -0.0025 ($p = 0.48$): the estimate is not an artifact of how heavily that one bloc weights the aggregation. Re-including Hong Kong and Macao gives -0.0009 ($p = 0.80$). Restricting to common-support products gives -0.0022 ($p = 0.57$). Across every variation the dirty coefficient sits between -0.0040 and -0.0060 with asymptotic p-values between 0.02 and 0.09 — persistently marginal, and, as Section 4.4 established, robust to none of the bootstrap, permutation, or leave-one-out checks.

Table 10: Sample robustness (full firm-level panel, WB index)

Variation	Obs.	EP×green (p)	EP×dirty (p)
Baseline	21.5M	-0.0023 (0.57)	-0.0044 (0.052)
With controls	19.3M	-0.0002 (0.94)	-0.0060 (0.022)
Excluding ASEAN	19.2M	-0.0025 (0.48)	-0.0044 (0.051)
Including HK–Macao	23.6M	-0.0009 (0.80)	-0.0041 (0.094)
Common support	21.5M	-0.0022 (0.57)	-0.0044 (0.052)
PPML, zero-filled grid	7.9M cells	$+0.0015$ (0.74)	-0.0301 (0.06)
Within-firm green share	13.3M	-0.0001 (0.50)	

Asymptotic destination-clustered p-values in parentheses; the dirty column’s marginal significance does not survive wild cluster bootstrap, permutation, or leave-one-out inference (Section 4.4). Controls: $\ln(1+\text{MFN tariff})$, $\ln \text{HHI}$, antidumping indicator.

A remaining caveat concerns the tariff control, and it is worth stating with the provenance made explicit. The variable is the applied duty recorded in the customs data rather than a rate constructed here; it varies across destinations for the same HS6–year in 97.9% of product–year cells, so it is a destination-level tariff facing Chinese exporters and not a Chinese import duty, and its mean does not fall after PTA entry for partner destinations, so it is the MFN rather than the bilateral preferential rate. The preferential rate was not obtainable (the WITS-TRAINS API was unavailable at the time of writing). Three considerations bound what this costs. The tariff does not enter equation (1) at all — it appears only in the “with controls” row of Table 10 and in the CEM matching — so the headline estimates

do not depend on it. Whatever tariff liberalization the agreements delivered varies at the destination–year level and is absorbed by θ_{dt} , and at the product level by θ_{pt} . And the residual threat runs in a conservative direction: if PTA tariff cuts had been deeper on green goods than on neutral ones, the omitted preferential margin would load onto $EP \times green$ with a positive sign, inflating rather than suppressing the estimate. The estimate is zero.

5.7 Destination-specific trends in the composition gap

The conservative-bias argument of Section 3 has a testable counterpart, and it is the paper’s central falsification exercise: if destinations signing deep environmental chapters have green demand rising for reasons of their own — a rival explanation for the null result, not merely a robustness formality — allowing each destination its own linear trend in the green (and dirty) gap, estimated from data the treatment cannot have contaminated, should matter. Two variants of this exercise, run on the collapsed panel, bracket the answer. The first adds destination-specific linear trends interacted with green and dirty status, estimated over the full sample period. The WB results are essentially unchanged ($EP \times green$ -0.0051 , bootstrap $p = 0.50$; $EP \times dirty$ -0.0082 , bootstrap $p = 0.28$), but the TREND green interaction flips sign relative to the baseline (from $+0.0018$, $p = 0.32$, to -0.0022) and — the only coefficient in the paper’s destination-trend and robustness battery to do so — survives the wild cluster bootstrap ($p = 0.012$). Taken at face value it would say environmental provisions *reduce* green exports. It should not be taken at face value: unit-specific trends estimated over the full sample are known to absorb post-treatment dynamics and can reverse the sign of treatment coefficients when effects or cohort dynamics are present (Wolfers, 2006), and the negative late drift of the green gap that these trends pick up is precisely the early-cohort artifact that the Sun–Abraham estimator already dismissed (Appendix A). The second variant removes the contamination by construction: destination-specific trends are estimated on *pre-treatment* years only (all years for never-treated destinations) and projected forward before re-estimating the main specification on the detrended outcome. Every coefficient returns to an imprecise zero. (That variant estimates destination slopes in a first stage and subtracts them before re-estimating, so the second-stage inference does not carry the first-stage estimation error; with intervals this wide the omission is formal rather than material, but it is one more reason to read the exercise as diagnostic.) The coefficients are $TREND \times green$ $+0.0074$, bootstrap $p = 0.18$; $WB \times green$ $+0.0168$, bootstrap $p = 0.71$; dirty interactions $p = 0.30$ – 0.62). That the one nominally robust coefficient in the trend specification reverses sign entirely depending on how the trends are estimated identifies it as an artifact of the trend specification, not a treatment effect; the composition null stands against destination-specific linear drifts in green demand, the most plausible form of the selection-on-trends confounder.

5.8 A continuous dirty margin: CO₂ intensity

The binary dirty indicator groups HS6 products into the six classic pollution-intensive sectors of Mani & Wheeler (1998); a natural concern is that a continuous, externally sourced measure of pollution intensity might tell a different story. I construct one from the replication data of Shapiro (2021), which reports China-specific CO₂ emission intensity (direct and total, i.e. including upstream inputs) for 47 EXIOBASE industries. Each industry code encodes a two-digit ISIC Rev. 3 / NACE Rev. 1.1 division (verified against the subset of industries the replication package labels by name: e.g. *Pulp* loads on division 21, *Coke oven products* on division 23, *Cement, lime and plaster* on division 26, *Iron and steel* and *Aluminum* on division 27 — exactly the ISIC divisions underlying the six Mani-Wheeler sectors). HS6 products are mapped to ISIC3 divisions via the WITS HS1996–ISIC3 concordance already used for the binary classification, and assigned the simple average China-specific intensity across the EXIOBASE sub-industries in their division (no sub-industry weights are available in the replication data); unmatched HS6 codes (9.5% of the panel) are assigned the sample mean. The resulting measure validates against the existing trichotomy without circularity, since it is built independently of the green/dirty/neutral classification: mean intensity is 0.0012 for dirty products, 0.0008 for neutral products, and 0.0006 for green products — monotonic in the expected direction.

Re-estimating equation (1) on the collapsed panel with EP interacted with the standardized continuous intensity in place of the binary dirty indicator gives the same reading as the rest of the paper: EP×intensity is -0.0025 with the WB index (bootstrap $p = 0.71$) and -0.0013 with TREND (asymptotic $p = 0.012$, bootstrap $p = 0.06$) — a fade under robust inference consistent with the marginal, non-robust pattern documented throughout Section 4.4, not a new or sharper result. The continuous measure does not overturn the conclusion that the composition null is not an artifact of how dirty products are classified.

6 Conclusion

Using the universe of Chinese customs transactions and two independent codings of the environmental content of China’s trade agreements, this paper finds that environmental provisions did not change what China exports on the green margin or the extensive margin: a null that is stable across estimation designs and sample variations and that survives an inference battery designed for the few-treated-clusters regime characterizing this — and most — PTA settings. Its bootstrap bounds rule out effects larger than about one quarter of the benchmark estimate in the aggregate literature; below that threshold the design does not discriminate, and the claim is bounded rather than exact. A descriptive within-firm share

panel points the same way, though the level specification it requires is not separately identified here (Section 5). The dirty margin is the one apparent exception, and it does not survive scrutiny: a negative effect on dirty exports that is overwhelming under asymptotic standard errors clears none of the same battery and is driven by a single destination (Section 4.4) — a false positive of the kind few-treated-cluster designs manufacture, reported here as a descriptive pattern and a cautionary case rather than an identified effect.

The reading is not that environmental provisions never matter: Abman et al. (2024) show they can, on environmental outcomes, when they are specific and enforceable. The reading is that the environmental chapters China signed in 2000–2015 — thin, bundled, and dominated by cooperation language — had no trade bite. For the policy debate on greening trade agreements, the distinction matters: content design, not chapter presence, is what could move flows. For the empirical literature, the paper adds a cautionary methodological tale: with treatment varying across a dozen agreements, asymptotically significant composition effects appear and dissolve under honest inference; the discipline of bootstrap, permutation, and leave-one-out should be standard equipment.

References

- Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1), 1–35.
- Abman, R., Lundberg, C., & Ruta, M. (2024). The effectiveness of environmental provisions in regional trade agreements. *Journal of the European Economic Association*, 22(6), 2507–2548.
- Baghdadi, L., Martinez-Zarzoso, I., & Zitouna, H. (2013). Are RTA agreements with environmental provisions reducing emissions? *Journal of International Economics*, 90(2), 378–390.
- Baccini, L., Pinto, P. M., & Weymouth, S. (2017). The distributional consequences of preferential trade liberalization: firm-level evidence. *International Organization*, 71(2), 373–395.
- Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1), 249–275.
- Brandi, C., Schwab, J., Berger, A., & Morin, J.-F. (2020). Do environmental provisions in trade agreements make exports from developing countries greener? *World Development*, 129, 104899.

- Callaway, B., & Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.
- Callaway, B., Goodman-Bacon, A., & Sant’Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. NBER Working Paper No. 32117.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.
- Cherniwchan, J. (2017). Trade liberalization and the environment: evidence from NAFTA and U.S. manufacturing. *Journal of International Economics*, 105, 130–149.
- Conley, T. G., & Taber, C. R. (2011). Inference with “difference in differences” with a small number of policy changes. *Review of Economics and Statistics*, 93(1), 113–125.
- Copeland, B. R., Shapiro, J. S., & Taylor, M. S. (2022). Globalization and the environment. In *Handbook of International Economics* (Vol. 5, pp. 61–146). Elsevier.
- Correia, S. (2017). Linear models with high-dimensional fixed effects: an efficient and feasible estimator. Working paper.
- de Chaisemartin, C., & D’Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964–2996.
- Dechezleprêtre, A., & Sato, M. (2017). The impacts of environmental regulations on competitiveness. *Review of Environmental Economics and Policy*, 11(2), 183–206.
- Dür, A., Baccini, L., & Elsig, M. (2014). The design of international trade agreements: introducing a new dataset. *Review of International Organizations*, 9(3), 353–375.
- Feenstra, R. C., & Hanson, G. H. (2004). Intermediaries in entrepôt trade: Hong Kong re-exports of Chinese goods. *Journal of Economics & Management Strategy*, 13(1), 3–35.
- Fischer, A., & Roodman, D. (2021). fwildclusterboot: fast wild cluster bootstrap inference for linear regression models. R package.
- Fisher, R. A. (1935). *The Design of Experiments*. Oliver & Boyd.
- Eckel, C., & Neary, J. P. (2010). Multi-product firms and flexible manufacturing in the global economy. *Review of Economic Studies*, 77(1), 188–217.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.

- Head, K., & Mayer, T. (2014). Gravity equations: workhorse, toolkit, and cookbook. In *Handbook of International Economics* (Vol. 4, pp. 131–195). Elsevier.
- Hofmann, C., Osnago, A., & Ruta, M. (2017). Horizontal depth: a new database on the content of preferential trade agreements. *World Bank Policy Research Working Paper*, 7981.
- Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: coarsened exact matching. *Political Analysis*, 20(1), 1–24.
- Larch, M., Shikher, S., & Yotov, Y. V. (2025). Estimating gravity equations: theory implications, econometric developments, and practical recommendations. *Review of International Economics*, 33(5), 1066–1092.
- Low, P., & Yeats, A. (1992). Do “dirty” industries migrate? In P. Low (ed.), *International Trade and the Environment*, World Bank Discussion Paper 159.
- MacKinnon, J. G., & Webb, M. D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2), 233–254.
- Mani, M., & Wheeler, D. (1998). In search of pollution havens? Dirty industry in the world economy, 1960–1995. *Journal of Environment & Development*, 7(3), 215–247.
- Morin, J.-F., Dür, A., & Lechner, L. (2018). Mapping the trade and environment nexus: insights from a new dataset. *Global Environmental Politics*, 18(1), 122–139.
- Neri-Lainé, M., Orefice, G., & Ruta, M. (2023). Deep trade agreements and heterogeneous firms’ exports. *CESifo Working Paper*, 10436.
- Rajan, R. G., & Zingales, L. (1998). Financial dependence and growth. *American Economic Review*, 88(3), 559–586.
- Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., & Webb, M. D. (2019). Fast and wild: bootstrap inference in Stata using boottest. *Stata Journal*, 19(1), 4–60.
- Santos Silva, J. M. C., & Tenreyro, S. (2006). The log of gravity. *Review of Economics and Statistics*, 88(4), 641–658.
- Sauvage, J. (2014). The stringency of environmental regulations and trade in environmental goods. *OECD Trade and Environment Working Papers*, No. 2014/03, OECD Publishing.
- Shapiro, J. S. (2021). The environmental bias of trade policy. *Quarterly Journal of Economics*, 136(2), 831–886.

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199.

Wolfers, J. (2006). Did unilateral divorce laws raise divorce rates? A reconciliation and new results. *American Economic Review*, 96(5), 1802–1820.

A Sun–Abraham dynamics on the composition gap

Figure 2 plots the interaction-weighted event study of Sun & Abraham (2021) estimated on the destination–year composition gaps (the difference between the weighted mean outcome of green, respectively dirty, products and that of neutral products within each destination–year), with never-treated destinations as the control group and $t = -1$ as the reference period. The aggregated ATTs confirm the static results: -0.044 ($p = 0.24$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap.

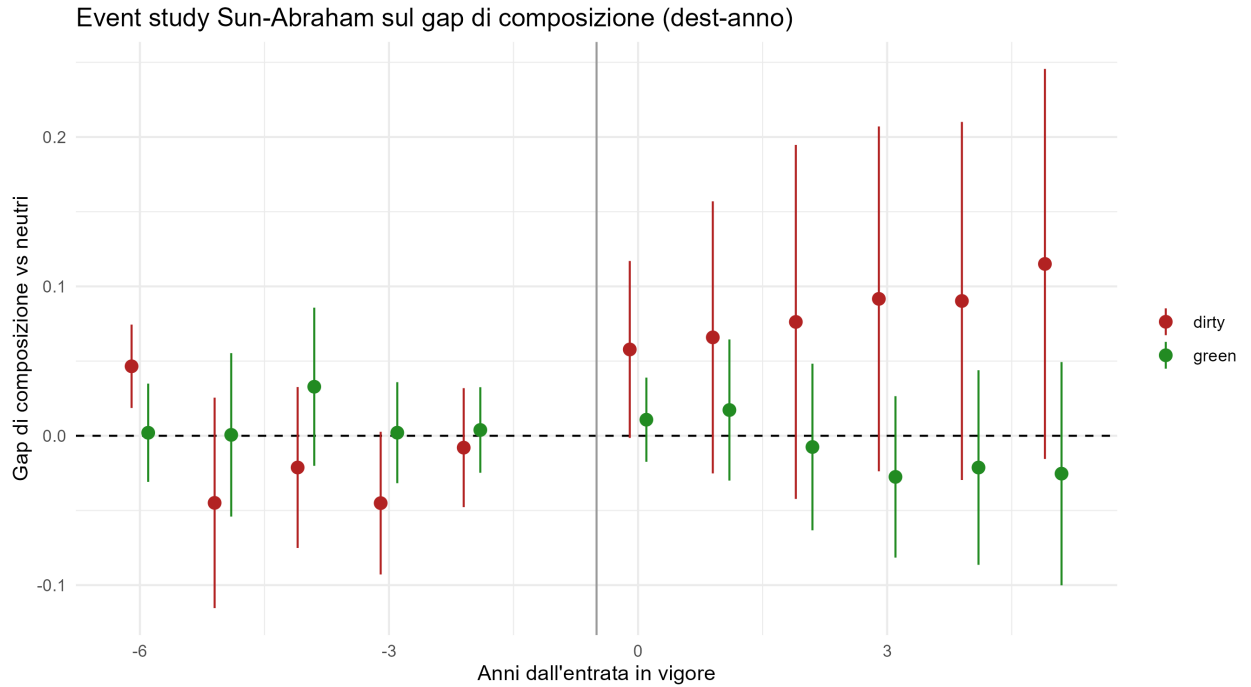


Figure 2: Sun–Abraham event study on the destination–year composition gap, window $[-6, +5]$; destination and year fixed effects, cell-count weights, destination-clustered 95% bands; $t = -1$ is the reference period.

One coefficient in the pre-period deserves explicit discussion: the dirty gap at $t = -6$ is $+0.047$ with an asymptotic p -value of 0.001. Taken at face value it would qualify the

flat-pre-trend reading of Section 4. A cohort decomposition shows it should not be taken at face value, for four reasons.

First, distant leads are identified by very few units: the sample starts in 2000, so $t = -6$ exists only for the eight destinations that entered in 2006 or later (Chile, Pakistan, New Zealand, Peru, Costa Rica, Iceland, Switzerland, Australia); deeper leads rest on as few as a single destination.

Second, the cohort-specific estimates underlying the aggregate are mutually contradictory: at $t = -6$ the 2006 cohort (Chile) deviates by -0.49 and the 2010 cohort (Peru) by -0.33 , against $+0.14$ for the 2008 cohort (New Zealand) and $+0.10$ for the 2015 cohort (Australia). There is no common displacement of dirty exports ahead of the agreements — only destination-specific fluctuations of opposite sign, averaged into a single number with trade-size weights.

Third, the aggregate is not robust to any single cohort: dropping the 2015 cohort (Australia alone, whose $t = -6$ falls in 2009) flips the coefficient to -0.051 ($p = 0.11$); dropping the 2014–15 cohorts yields -0.109 . A genuine pre-trend does not change sign when one destination is removed. Nor does it appear only at one isolated lead: the adjacent leads $t = -5$ through $t = -2$, where 17 to 23 destinations are observed, are all small and statistically indistinguishable from zero, and the two-way fixed-effects event study of Figure 1 shows nothing at $t = -6$ either ($+0.027$, $p = 0.53$).

Fourth, the design estimates roughly 28 lead and lag coefficients with 23 treated clusters, so the clustered variance matrix is rank-deficient and individual-lead p -values are unreliable by construction; with this many coefficients, one or two nominally significant leads are what sampling noise delivers under exactly flat pre-trends. The aggregated ATT — the statistic with a valid interpretation — remains null in every variant: $p = 0.27$ in the baseline, $p = 0.11$ restricting to the $[-6, +5]$ window, $p = 0.29$ excluding the 2014–15 cohorts.

B Collapsed-panel and full-panel equivalence

Table 11 compares the collapsed-panel estimates (R, cell-count weighted, FE $pd+dt+pt$) with the full firm-level panel estimated under the same fixed-effect structure (Stata, unweighted, FE $pd+dt+pt$). The two rows agree to seven significant figures, confirming that the weighted collapsed regression is algebraically identical to the unweighted micro regression under the same fixed effects.

Table 11: Collapsed vs. full panel under $pd + dt + pt$ fixed effects (WB index)

	EP $\times$ green	EP $\times$ dirty	Obs.
Collapsed panel ($pd + dt + pt$, R)	−0.0046 (0.0070)	−0.0119 (0.0030)	3.7M
Full panel ($pd + dt + pt$, Stata)	−0.0046 (0.0069)	−0.0119 (0.0029)	45.7M

WB index. Destination-clustered standard errors in parentheses. Both rows use FE $pd + dt + pt$. The collapsed-panel estimates (R, cell-count weighted) and the full-panel estimates (Stata, unweighted) agree to seven significant figures, confirming the algebraic equivalence. The full panel retains the firm dimension.